

Anish Acharya, PhD

Senior Research Scientist
Amazon Web Services (AWS)

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Website • Scholar • GitHub • LinkedIn

Education

- **The University of Texas at Austin**
PhD, Electrical and Computer Engineering
Advisor(s): **Prof. Inderjit S. Dhillon** and **Prof. Sujay Sanghavi**
PhD Committee: **Prof. Alex Dimakis**, **Prof. Ludwig Schmidt**, and **Prof. Nhat Ho**
PhD Thesis: **Robust and Efficient Learning in High Dimensions from Noisy Data**
Austin, TX
Fall, 2019 – Spring, 2025
- **University of California, Irvine**
M.S., Electrical Engineering and Computer Science
Irvine, CA
Fall, 2013 – Fall, 2014
- **Jadavpur University**
B.E., Electronics and Instrumentation Engineering
Advisor(s): **Amitava Gupta** and **Shantanu Das**
Kolkata, India
Spring, 2009 – Spring, 2013

Research Interests

My recent focus is **Recursive Self Improvement (RSI)**: the loop in which an agentic system proposes changes to its own policy (prompts, tools, orchestration logic and weights), scores them, and keeps what scored better. I work on both halves of that loop, the **signal** it optimizes against and the **search** that exploits it, together with the theory of when iterating provably helps. More broadly: **robust reward modelling and evaluation** under biased or adversarial supervision, **multi-agent systems**, **constrained combinatorial search**, and **robust, distributed and federated optimization**.

Research Experience

- **Amazon**
Senior Research Scientist, AWS (Full Time)
Building Decentralized AI Marketplace.
Lead the full life-cycle of AWS **multi-agent orchestration meshes** – architecture, core research, production integration, and eval-driven iteration. Research on **recursive self improvement** – robust reward modeling (**RoPoLL**), prompt optimization(**SPLICE**), and orchestration search (**OrchOpt**) – directly powers: **AI-assisted Product Listing**, AWS Summit NY 2026 (**blog, docs, video**); **AWS Partner Central Agents (onboarding, funding, MCP)**; **AWS Marketplace Agent Mode**, re:Invent 2025 (**blog, product, ZDNet, CRN, The Ravit Show, Inside Partnering**).
Austin, TX
Spring, 2025 – Present
- **University of Texas at Austin**
Graduate Research Assistant (Part Time)
*Doctoral research on **robust, sample-efficient learning in high dimensions**: robust estimation under label noise and Byzantine corruption, distributed and federated optimization, and data-efficient subset selection (**thesis**). Affiliated with the **Institute for Foundations of Machine Learning (IFML)**.*
Austin, TX
Fall, 2019 – Spring, 2025
- **Eidon AI**
Head of AI Research (Part Time)
The AI is the Blockchain, the Blockchain is the AI.
Architected and launched a **multi-agent dynamic point-based system** integrating blockchain with AI: evaluation metrics for content quality, AI alignment, and community standards, and a **tokenomics framework for reward distribution** across the decentralized network (**Technical Docs**).
Austin, TX
Summer, 2024 – Winter, 2025
- **Meta**
Student Researcher (Part Time)
Research Scientist (Intern)
Student Researcher (Part Time)
Research Scientist (Intern)
*With Meta Capacity Engineering and Meta Fundamental AI Research (FAIR), hosted by **Li Jing**, **Bhargav Bhushanam**, **Dhruv Choudhary**, and **Michael Rabbat**: online, low-resource training to scale Facebook’s deep recommendation models. Developed **positive-unlabeled contrastive representation learning** for weakly supervised click-through-rate prediction, learning from positives and unlabeled data without negative sampling – **ICML 2022 PODS workshop (paper)** and **extended version**.*
Menlo Park, CA
Fall, 2022
Summer, 2022
Fall, 2021
Summer, 2021
- **Microsoft**
Research Scientist (Intern)
*As part of the **Speech and Dialog Research Group** (hosted by **Kenichi Kumatani** and **Dimitrios Dimitriadis**), developed algorithms*
Redmond, WA
Summer, 2020

for training large-scale models robust to **adversarial, Byzantine, and noisy client updates** in the federated setting; built **OpenMSFTL**, an open-source federated learning platform ([github](#)).

- **Amazon** **Sunnyvale, CA**
Applied Scientist, AGI (Full Time) *Summer, 2018 – Spring, 2019*
Applied Scientist, Alexa AI (Full Time) *Spring, 2018 – Summer, 2018*
Applied Scientist, Alexa ML (Full Time) *Winter, 2017 – Spring, 2018*
Software Development Engineer, Amazon Devices (Full Time) *Summer, 2016 – Winter, 2017*
Shipped NLP domain models shared across Alexa devices – Timers, Alarms, Reminders, Calendar, Communications – on Echo Show, Fire TV, and other Alexa IoT devices. **Primary inventor on: Alexa Conversations**, goal-oriented dialog generation from dialog templates, APIs, and entity data ([paper](#), [patent](#), [skill](#), [Amazon Science](#), [CNN](#), [Forbes](#), [VentureBeat](#)); **online embedding compression for deployment-constrained NLU** – low-rank factorization folded into training, so task data shapes the compression itself, deployed across a large fleet of per-skill models ([paper](#), [patent](#), [Amazon Science](#), [Amazon Blog](#), [Packt](#), [Hacker News](#)).
- **eBay Inc.** **San Jose, CA**
Data Scientist, Customer Insights and Analytics (Contract – Manthan Systems Inc.) *Fall, 2015 – Summer, 2016*
Developed **deep recommendation algorithms** to personalize buyer search and targeted advertising, e.g., **generative models for personalized emails**.
- **Schlumberger** **Houston, TX**
Data Scientist (Full Time) *Spring, 2015 – Fall, 2015*
Early member of the **Digital Transformation Services** AI research group: **failure prediction** for deployed oil service instruments and **oil level estimation** from extraction life-cycle sensor data.
- **FEM Inc.** (acquired by **Gracenote**, a Nielsen company) **Los Angeles, CA**
Data Engineer (Full Time) *Winter, 2015*
With the co-founders: NLP models for a content recommendation system, and visualization tooling for an interactive analytics dashboard.
- **Toyota Technological Institute Chicago (TTIC)** **Chicago, IL; Ann Arbor, MI**
Visiting Research Scientist (Intern) *Summer, 2014*
Hosted by **Prof. David McAllester**: real-time stereo algorithm for 3D scene reconstruction for autonomous vehicles, plus a stereo evaluation benchmark based on a slanted-plane scene model; built for **Toyota Research in North America (TRINA)**.

Teaching Experience

- **Amazon Web Services** **Austin, TX**
Instructor
AWS GenAI Training: **Embedding AI into Products: Evaluation-Driven Development** *Summer, 2026*
AWS GenAI Training: **Embedding AI into Products: Governance and Guardrails** *Spring, 2026*
- **University of Texas at Austin** **Austin, TX**
Instructor *Fall, 2025*
Directed Reading Program (DiRP): **Natural Language Optimization (NLO) for LLMs**
Teaching Assistant
MIS 285N: **Introduction to Deep Learning**, Instructors: Prof. C. Caramanis, Prof. A. Dimakis *Spring, 2024*
MIS 185N: **Emerging Trends in Machine Learning**, Instructor: Prof. S. Shakkottai *Spring, 2024*
ECE 461P: **Data Science Principles**, Instructor: Prof. S. Sanghavi *Fall, 2023*
- **University of California Irvine** **Irvine, CA**
Teaching Assistant
EECS 50: **Discrete Time Signals and Systems** (Undergraduate) *Fall, 2014*
EECS 280A: **Advanced Engineering Electromagnetics I** (Graduate) *Spring, 2014*
EECS 118: **Introduction to Knowledge Management and Software Engineering** (Undergraduate) *Winter, 2014*
EECS 119: **Introduction to VLSI** (Undergraduate) *Fall, 2013*

Professional Service

- **Reviewer, Conferences:**
 - **International Conference on Machine Learning (ICML)**
 - **Conference on Neural Information Processing Systems (NeurIPS)**
 - **International Conference on Learning Representations (ICLR)**
 - **International Conference on Artificial Intelligence and Statistics (AISTATS)**
 - **Conference on Uncertainty in Artificial Intelligence (UAI)**
 - **Annual Meeting of the Association for Computational Linguistics (ACL)**
 - **IEEE International Symposium on Information Theory (ISIT)**

- **Reviewer, Journals:**
 - [Journal of Machine Learning Research](#) (MIT Press)
 - [Transactions on Knowledge and Data Engineering](#) (IEEE)
 - [Pattern Recognition Letters](#) (Elsevier)
 - [Cognitive Neurodynamics](#) (Springer Nature)
 - [Discover Electronics](#) (Springer Nature)
- **Program Committee:**
 - [International Conference on Machine Learning \(ICML\)](#)
 - [AAAI Conference on Artificial Intelligence \(AAAI\)](#)

Honors and Awards

- **Silver Reviewer** award at ICML 2026. Spring, 2026.
 - **Travel Grants**, UT Austin – ML conferences (NeurIPS, ICML, AISTATS, UAI) and Simons Institute, UC Berkeley. 2021 – 2024.
 - **Meta PhD Research Grant** of \$100,000 Spring, 2021 – Spring, 2023.
 - **Meta PhD Research Fellowship Finalist** – AI HW/SW co-design ([Press Release](#)) Spring, 2022.
 - Awarded Bachelor of Engineering with **Highest Honors** (First Class), Jadavpur University Fall, 2013
 - **INSPIRE Fellowship** and **MHRD Fellowship for Academic Excellence**, Govt. of India. Fall, 2009 – Spring, 2013.
 - **State Rank 5** of ~700,000 candidates, West Bengal Board of Secondary Education ([The Telegraph](#)). Spring, 2007
 - **Mamraj Agarwal Rashtriya Puraskar** (National Award for Academic Excellence), from the **Governor of West Bengal**. Summer, 2007
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Granted US Patents (Primary Inventor)

1. (Amazon) **Goal Oriented Dialog Generation using Dialog Template, API and Entity data**; July, 2022. [USPTO](#)
A. Acharya, A. Metallinou, T. Chung, S. Paul, S. Chandra, C. Lin, D. H. Tur, A. Mandal
 2. (Amazon) **Natural Language Processing: Amazon Alexa AI**; December, 2020. [USPTO](#)
A. Acharya, A. Metallinou, R. Goel, I. Dhillon
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Publications

1. **SPLICE: Structured Prompt Local Iterative Combinatorial Evolution.**
A. Acharya, P. Studans, A. Dhanda, N. V. Rao, V. A. Khatri, S. A. Niaki, B. Verkhovsky.
Conference on Neural Information Processing Systems (NeurIPS), 2026
2. **RoPoLL: Robust Panel of LLM Judges.**
A. Acharya, K. Pan, B. Verkhovsky.
International Conference on Machine Learning (ICML), 2026
Workshop on Statistical Frameworks for Uncertainty in Agentic Systems (AgenticUQ)
• [OpenReview](#) • [Arxiv](#) • [Amazon Science](#)
3. **OrchOpt: Validation-Gated Architecture Search for Self-Organizing Multi-Agent Orchestration.**
A. Dhanda, S. A. Niaki, A. Acharya, B. Verkhovsky.
Conference on Language Modeling (COLM), 2026
2nd Workshop on Lifelong Agents: Learning, Aligning, and Evolving
• [OpenReview](#) • [Amazon Science](#)
4. **Geometric Median Matching for Robust k-Subset Selection from Noisy Data.**
A. Acharya, S. Sanghavi, A. G. Dimakis, I. S. Dhillon.
International Conference on Machine Learning (ICML), 2025
• [PMLR](#) • [OpenReview](#) • [Arxiv](#)
5. **Neural Distributed Source Coding.**
A. Nagle, J. Whang, A. Acharya, H. Kim, A. Dimakis.
IEEE Journal on Selected Areas in Information Theory, 2024
• [IEEE Xplore](#) • [Arxiv](#)
6. **Faster Non-Convex Federated Learning via Global and Local Momentum.**
R. Das, A. Acharya, A. Hashemi, S. Sanghavi, I. Dhillon, U. Topcu.
The Conference on Uncertainty in Artificial Intelligence (UAI), 2022
• [OpenReview](#)

SPOTLIGHT

7. **Robust Training in High Dimensions via Block Coordinate Geometric Median Descent.**

A. Acharya, A. Hashemi, P. Jain, S. Sanghavi, I. Dhillon, U. Topcu.

International Conference on Artificial Intelligence and Statistics (AISTATS), 2022

• [PMLR](#)

8. **Positive Unlabeled Contrastive Learning.**

A. Acharya, S. Sanghavi, L. Jing, B. Bhushanam, D. Choudhary, M. Rabbat, I. Dhillon.

International Conference on Machine Learning (ICML), 2022

Principles of Distribution Shift (PODS) Workshop

• [Proceedings](#) • [Arxiv](#)

9. **LDKP: A Dataset for Identifying Keyphrases from Long Scientific Documents.**

D. Mahata, N. Agarwal, D. Gautam, A. Kumar, S. Parekh, Y. K. Singla, A. Acharya, R. R. Shah.

ACM International Conference on Information and Knowledge Management (CIKM), 2022

DL4SR'22: Workshop on Deep Learning for Search and Recommendation

• [CEUR-WS](#) • [Arxiv](#) • [HF Dataset](#)

10. **Alexa Conversations: An Extensible Data-driven Approach for Building Task-oriented Dialogue Systems.**

DEMO

A. Acharya, S. Adhikari, S. Agarwal, V. Auvray, N. Belgamwar, A. Biswas, S. Chandra, T. Chung, M. Zarandi, R. Gabriel, S. Gao, R. Goel, D. Hakkani-Tur, J. Jezabek, A. Jha, J. Kao, P. Krishnan, P. Ku, A. Goyal, C. Lin, Q. Liu, A. Mandal, A. Metallinou, V. Naik, Y. Pan, S. Paul, V. Perera, A. Sethi, M. Shen, N. Strom and E. Wang.

Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021

• [ACLAnthology](#)

11. **GupShup: An Annotated Corpus for Abstractive Summarization of Open-Domain Code-Switched Conversations.**

L. Mehnaz, D. Mahata, A. Kumar, U. S. Gunturi, R. Jain, G. Gupta, R. Gosangi, I. G. Lee, A. Acharya, R. Shah.

Conference on Empirical Methods in Natural Language Processing (EMNLP), 2021

• [ACLAnthology](#)

12. **On the Benefits of Multiple Gossip Steps in Communication Constrained Federated Learning.**

A. Hashemi, A. Acharya, R. Das, H. Vikalo, S. Sanghavi, I. Dhillon.

IEEE Transactions on Parallel and Distributed Systems, 2021

• [IEEE Xplore](#)

13. **Online Embedding Compression for Text Classification using Low Rank Matrix Factorization.**

SPOTLIGHT

A. Acharya, R. Goel, A. Metallinou, I. Dhillon.

AAAI Conference on Artificial Intelligence (AAAI), 2019

• [Proceedings](#)

14. **Extending The Concept of Analog Butterworth Filter For Fractional Domain.**

A. Acharya, S. Das, I. Pan, S. Das.

Signal Processing (Elsevier), 2014

• [ScienceDirect](#) • [Arxiv](#)

15. **Simulation studies on the design of optimum PID controllers to suppress chaotic oscillations in a family of Lorenz-like multi-wing attractors.**

S. Das, A. Acharya, I. Pan.

Mathematics and Computers in Simulation (Elsevier), 2014

• [ScienceDirect](#) • [Arxiv](#)

16. **Stability Analysis Of Delayed System Using Bode's Integral.**

ORAL

A. Acharya, D. Mitra, K. Halder.

International Conference on Computer Communication and Informatics (ICCCI), 2013

• [IEEE Xplore](#)

17. **Optimum PID Control of Multi-wing Attractors in A Family of Lorenz-like Chaotic Systems.**

ORAL

A. Acharya, S. Das, I. Pan.

International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2012

• [IEEE Xplore](#)

18. **Optimized Quality Factor of Fractional Order Analog Filters with Band-Pass and Band-Stop Characteristics.**

ORAL

A. Pakhira, S. Das, A. Acharya, I. Pan, S. Saha.

International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2012

• [IEEE Xplore](#)

19. **Identification of nonlinear systems from the knowledge around different operating conditions: a feed-forward multilayer Artificial Neural Net (ANN) based approach.** ORAL
 S. Saha, S. Das, **A. Acharya**, A. Kumar, S. Mukherjee, I. Pan, A. Gupta.
International Conference on Parallel, Distributed and Grid Computing (PDGC), 2012
 • IEEE Xplore
20. **Least square and instrumental variable system identification of AC servo position control system with fractional Gaussian noise.** ORAL
 S. Das, A. Kumar, I. Pan, **A. Acharya**, S. Das, A. Gupta.
International Conference on Energy, Automation and Signal (ICEAS), 2011
 • IEEE Xplore

Invited Talks and Presentations

- Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Prompt Optimization (Oral)
 - Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Robust LLM Jury (Oral)
 - Summer, 2026. (Seoul, South Korea) **ICML 2026** – Robust LLM Jury (Poster)
 - Summer, 2026. (New York City, NY) **AWS Marketplace Science** – Multi-Agent Optimization (Invited Talk)
 - Summer, 2026. (Virtual) **AI Across AWS** – Agent Optimization (Invited Talk)
 - Spring, 2026. (Virtual) **AWS Town Hall** – Prompt Optimization (Invited Talk)
 - Fall, 2025. (Seattle, WA) **Amazon Fulfillment Simulation Symposium (AFSS)** – Robust Sampling (Invited Talk)
 - Summer, 2025. (Vancouver, Canada) **ICML 2025** – Robust Sampling (Recorded Talk, Poster)
 - Fall, 2023. (Austin, TX) **Future of Chatbots and Conversational AI Summit (FOCA 2023)** (Keynote, Panel)
 - Fall, 2022. (Virtual) **Meta: Fundamental AI Research (FAIR)** (Invited Talk)
 - Fall, 2022. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Optimization (Poster)
 - Fall, 2022. (Virtual) **Meta: Ads ML Research** (Invited Talk)
 - Summer, 2022. (Eindhoven, Netherlands) **UAI 2022** – Faster Non-Convex Federated Learning (Poster)
 - Summer, 2022. (Baltimore, MD) **ICML 2022** – Robust Representation Learning (Recorded Talk, Poster)
 - Spring, 2022. (Austin, TX) **6g@UT Forum** – Robust Federated Learning (Poster)
 - Spring, 2022. (Virtual) **AISTATS 2022** – Robust Federated Learning (Recorded Talk, Poster)
 - Spring, 2022. (Virtual) **University of Exeter – Department of Mathematics** – Robust Federated Learning (Invited Talk)
 - Fall, 2021. (Virtual) **University of Denver – Department of Computer Science** – Robust Federated Learning (Guest Lecture)
 - Fall, 2021. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Federated Learning (Poster)
 - Fall, 2021. (Virtual) **Meta: Fundamental AI Research (FAIR)** – Robust Representation Learning (Invited Talk)
 - Summer, 2021. (Virtual) **Meta: Capacity Engineering Research** – Robust Representation Learning (Invited Talk)
 - Spring, 2021. (Virtual) **NSF TRIPODS** – Efficient Decentralized Optimization (Poster)
 - Summer, 2020. (Virtual) **Microsoft: Azure AI** – Robust Optimization (Invited Talk)
 - Fall, 2019. (Austin, TX) **UT Austin Graduate Research Showcase** – Efficient ML (Poster)
 - Fall, 2019. (Austin, TX) **Texas Wireless Summit** – Efficient ML (Poster)
 - Spring, 2019. (Honolulu, HI) **AAAI 2019** – Efficient ML (Oral)
 - Summer, 2018. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Efficient ML (Oral)
 - Winter, 2013. (Coimbatore, India) **IEEE ICCCI 2013** – Stability of Delayed Systems (Oral)
 - Fall, 2012. (Solan, India) **IEEE PDGC 2012** – Neural Network System Identification (Oral)
 - Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Chaos Control via Optimal PID (Oral)
 - Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Fractional Order Analog Filters (Oral)
 - Winter, 2011. (Bhubaneswar, India) **IEEE ICEAS 2011** – System Identification under Fractional Gaussian Noise (Oral)
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